

Assessing Redundant Interface Designs for Precise Number Input in Virtual Reality

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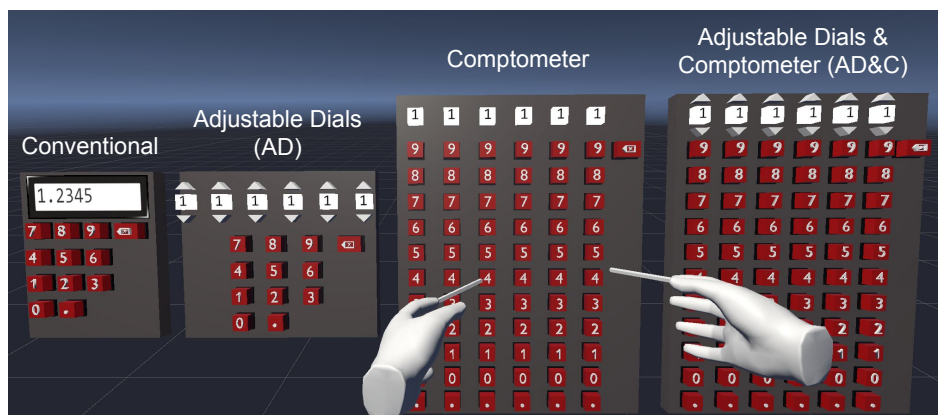


Figure 1: The conventional numpad is compared with three alternative designs that incorporate redundant interface and interaction elements to assist precise number input. These redundant features were inspired by analog mechanisms such as combination dials, movable numbers, and mechanical calculators with larger and multi-column number layouts (e.g., Comptometer [50]).

Abstract

Typing and editing precise numerical input, in particular floating-point values, are essential tasks in Spatial Computing for applications such as 3D precision modeling, object measurement, object dimensioning, mathematical visualization, and immersive media creation. Yet the adopted interfaces and interaction techniques in VR/AR/MR are often replicas of flat, palm-sized numpads such as those found in physical calculators or their WIMP counterparts. To move beyond such conventional confines, this paper explores redesigning the numpad by leveraging the spatial freedom of VR with a specific focus on introducing redundancy in the input of floating point values. To do so, we took inspiration from mechanisms such as combination dials, movable numbers, and a mechanical calculator

that offer a larger and multi-column number layout. We assess how redundant interfaces can enhance user experience and efficiency when it comes to precise number editing of floating point values. Through a user study (N=30), we compared four numpads where participants engage in inputting a list of target numbers within a virtual environment. Our findings reveal that the redesigned numpads, which utilize redundant design elements, were preferred by users over the conventional numpad design as they provided clearer and more efficient number input methods in VR.

CCS Concepts

• Computing methodologies → Motion processing; • Human-centered computing → Virtual reality; Ubiquitous and mobile computing.

Keywords

Virtual Reality, Input Redundancy, Numerical Keypads, User Study



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1 Introduction

When interacting with Spatial Computing interfaces, typing and editing precise numerical values is a recurrent task in scenarios where adjusting floating point numbers is a necessity [41, 45]. Examples include computer-aided design [37, 74], building information modeling [4, 11], parametric design [10], object measurement [46], object dimensioning [38], math function visualization [21], 3D content creation [1, 26, 61], 360°/180° video editing [58] or setting system control parameters [49].

Numpads (short for *numeric keypads*) allow users to type precise values of a specific number beforehand (i.e., fine parameter adjustment), thus acting as a complementary User Interface (UI) control element to sliders, which enable the exploration or expedite parameter control to find a serviceable numerical value (i.e., coarse parameter adjustment) [23, 23, 33, 34]. It is well known that whenever the precise numeric value matters, sliders are not the appropriate UI control element [23, 40].

In Virtual Reality (VR), numpads are often direct adaptations of Windows-Icons-Menus-Pointer (WIMP) interfaces or touchscreen designs, originally inspired by 1960s handheld calculators¹. These VR numpads feature number buttons (0-9) arranged in a grid layout on a small, floating panel. Several commercial VR applications such as Arkio [37], ShapesXR [12], Gravity Sketch [38], Layout [46] or even UnityStore assets [51, 55] currently use these conventional numpad designs (Fig. 2). One issue is that WIMP and touchscreen numpads were designed within very confined spatial restrictions (i.e., limited screen space). This design limitation is rooted in the metaphor originally derived from physical calculators, push-button telephones, or keypad entry locks that, due to physical constraints, offer limited interaction space. Another issue is that conventional VR numpads exhibit very limited interaction techniques, just allowing the user to type numbers via buttons and, seldom, to edit a number by placing the cursor at a designated decimal place.

As pointed out by Jason Jerald [28], designing novel widgets to input symbolic data in VR is essential for improving text and number entry. However, progress has been slow, largely because most VR widgets are mere replications of traditional desktop designs rather than being tailored to VR's unique affordances [25]. Such is the case of VR numpads. In VR, users are not limited by spatial restrictions, and can benefit from bi-manual interaction and body movements, which promote more natural input [6, 28]. While traditional numpad layouts emphasize compactness and limited interaction techniques, VR offers the opportunity for re-imagining this approach. Specifically, VR numpads can incorporate *redundancy*—offering multiple input options—to improve user clarity and input efficiency. Rather than being seen as restrictive, these redundancies, even if they require extra ‘real state’ and/or movement

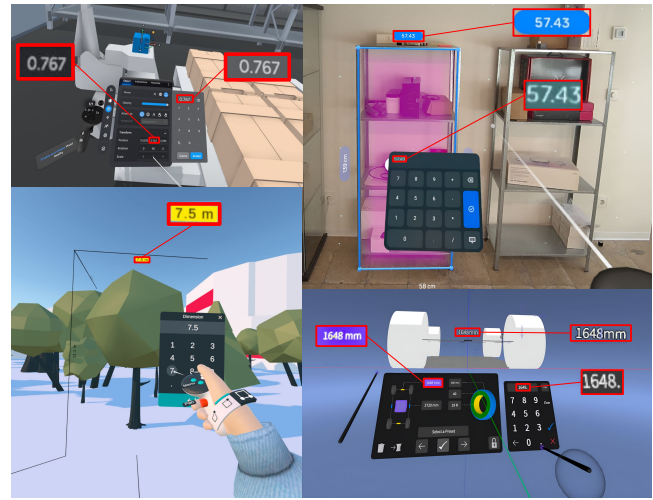


Figure 2: Numpads used for floating point number input in familiar XR apps: ShapesXR (top left), Layout (top right), Arkio (bottom left), and Gravity Sketch (bottom right).

leeway, could enhance user experience by offering more familiar, natural, and efficient ways to input numbers.

Previous work explored different interaction paradigms for text input within eXtended Reality (XR) environments [22, 47, 73]. Text and number input is a relevant task in various fields, including industrial simulations [19], medical and healthcare [39], education [57], scientific visualization [20, 48], financial and business training [31], gaming [63], or smart home and IoT systems [30]. Yet, to the best of our knowledge, there is a literature gap as numerical editing and input based on redundant interfaces remain unexplored.

Our work addresses this gap by redesigning VR numpad layouts, exploring the benefits of Spatial Computing and drawing inspiration from analog mechanisms such as combination dials found in padlocks, movable number blocks or mechanical calculators that offer larger and multi-column number layouts like the Comptometer². Without loss of generality, the proposed numpad redesigns focus on facilitating the entry and editing of six-character positive real numbers. We hypothesize that a numpad with built-in input redundancy and enhanced interaction techniques will outperform the conventional counterpart, enabling users to edit or input numbers more accurately, particularly for making small, frequent adjustments common in the iterative process of content creation, dimensioning or measurement [4, 10, 11, 26, 37, 38, 46, 58, 74].

Our study presents the following contributions: (i) we propose new VR numpad designs for precise number input and editing; (ii) we conducted a user study ($N = 30$) to evaluate the performance of three different VR numpads against the conventional VR numpad in terms of task completion time, Damerau-Levenshtein distances (i.e., number of changes required to transform one number into another), perceived usability, workload, user satisfaction, and user preference; finally, (iii) we demonstrate how redundant interface

¹<http://vintagecalculators.com/>

²<https://www.thecalculatorsite.com/articles/units/history-of-the-calculator.php>

and interaction techniques can improve user experience and efficiency in numerical input tasks, as well as our understanding on VR numpad design.

2 Related Work

Our work overlaps with the research question of how novel VR widgets for symbolic data input impact user performance [6, 9, 14, 27, 28]. In particular, we focus on how introducing redundancy in numpads can affect the input of floating point values. Although redundancy has been sparsely explored in multi-modal systems [56] and traditional WIMP interfaces [54, 66], its application to precise number input in XR remains largely unexamined, despite its potential to enhance clarity and efficiency [52].

When reviewing numpads in XR, it is essential to draw upon existing research concerning alphanumeric editing and input [3]. Text and numerical values are subsets of alphanumeric input (i.e., 26-letter Latin alphabet, numerical digits from 0-9, and special characters including @, #, and *). Clearly, the XR literature predominantly emphasizes text input, with only a handful of studies delving into numerical value input, particularly highlighting traditional numpad layouts.

Many studies have focused on text entry in XR environments [14] showcasing a diverse set of interfaces and range of interaction techniques to input alphanumeric values. These include eye gaze [42, 43], touchpad input [29], integration of machine learning to eliminate keyboards entirely [70], and modifications to common text entry metaphors to enhance user experience [2]. The spatial dimension offered by XR has also been explored to enhance text entry tasks, such as the design of novel virtual keyboards like the fanpad [68], cubic keyboard [69], keycube [8], or Punch Typing that investigates novel spatial gesture techniques based on key displacement [71]. These latter studies have inspired us to explore the spatial dimension offered by virtual environments, in particular, by transforming the numerical values placed on the display into 3D interactive elements (see Section 3.1) and by enlarging the widgets in comparison to their conventional counterparts. However, none of these studies truly focused on edition of floating point values *per se* as they predominantly dealt with text input, much less have such studies explored redundancy input for interface design.

Regarding VR studies that focus specifically on numpad design, the popular palm-sized grid layout (e.g., 3x3 or 4x3) remains predominant [5, 11, 35, 44]; with a few exceptions exploring radial layouts [49]. VR numpads have been studied in various contexts, including 3D modeling [11], PIN authentication [44], system control via button interaction [5, 9, 17], selection tasks [49], inputting mathematical expressions [57], and even money transaction [31]. As for VR numpad interaction techniques they predominantly involve single-finger touch or ray casting, with some studies exploring key displacement or drag-and-drop number input [57]. To this regard, Kreiser et al. developed the FloatOmeter, introducing the concept of presenting each digit of a number within its own designated box, facilitating faster and simpler modifications to individual digits [32]. Furthermore, Swieso et al. presented preliminary work on numpad input using machine learning to classify 3-D typing gestures on a phone numpad layout for digits 0 to 9, achieving just as good accuracy for number input [62]. Yet, just like the VR text entry

studies, none has considered redundancy input designs, nor have they compared such designs with the prevalent palm-sized numpad.

3 Numpads Widget Design

We developed numpad widgets to assist precise number input of floating point values by introducing redundancy in the interface design. Drawing inspiration from analog mechanisms (Fig.3), these widgets feature arrows for increasing or decreasing digit values (simulating the mechanics of combination padlock dials), displayed characters—such as numbers and decimal points—transformed into interactable *numboxes* that can be shifted along or detached from the display, and a multi-column button layout, inspired by mechanical calculators like the Comptometer.

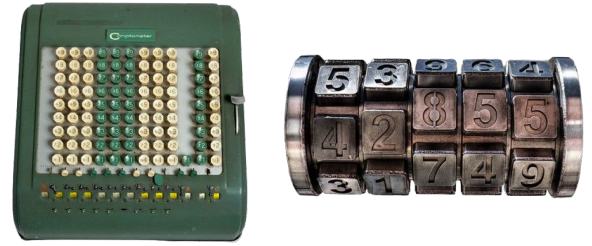


Figure 3: Analog mechanisms used as redundancy elements in our numpad designs: mechanical calculator (Comptometer from the 1950s) and combination dial mechanism.

A total of four numpad widgets were developed (Fig.1): the “Conventional” serves as our baseline, the “Adjustable Dials” (AD) introduces modular combination of dials and numboxes, the “Comptometer” replicates the multi-column button layout from mechanical calculators, and finally, the “Adjustable Dials & Comptometer” (AD&C) which combines numboxes, adjustable combination dials and multi-column button layout elements. Through this design approach, we aim to analyze the advantages and disadvantages of each numpad’s redundancies by assessing their performance, user satisfaction and preference. Note that these redundant input techniques are designed to improve precise number input or editing efficiency, particularly when making small numerical adjustments.

3.1 Interface and Interaction Elements

Pointer Interaction. Following the guidelines suggested by Speicher et al. [60], we chose to utilize a pointer as the means of interaction between the user and the numpad. This decision aims to reduce visual obstruction of the widget’s buttons caused by the interaction medium, especially given their small size during user engagement with the widget. The pointer functions as an extension of the user’s index finger, facilitating numerical input with either one or both hands across all of our numpad widgets, accommodating users’ preferences for single-handed or two-handed interaction.

Numpad Buttons. We chose three-dimensional square-shaped buttons, each with an area of 4 cm², to enhance input speed and accuracy while maintaining realism [15]. All numpads were placed within arm’s reach, ensuring users could access them with a simple extension of their arm. Table 1 details the dimensions of the

various user interface components. Additionally, upon pressing, these buttons exhibit a key-down animation to simulate real keys, providing a pseudo-haptic effect, accompanied by auditory feedback [59, 67]. We deliberately excluded actual haptic feedback, as the simultaneous use of visual, auditory, and haptic feedback may negatively impact task performance [16, 72].

Panel Tilting. To enhance the visualization of elements within the virtual environment, the numpads are inclined at a positive 15° angle, as they are positioned below the participants' line of sight. Conversely, the panels are inclined at a negative 30° angle, as they are situated above the participants' line of sight (Table 1).

Table 1: Dimensions of the UI components (all values in centimeters).

User Interface Component		Width	Height	Depth
Buttons	Number	Base	2.5	2.5
		Press	2	2
		Digit	-	-
	Delete	Base	3.8	2.5
		Press	3.3	2
		Backspace	-	-
	Increment		2	1
	Decrement		2	1
Conventional		20	25	2.5
AD		30	25	2.5
Comptometer and AD&C		37	50	2.5
Number Display Panel		50	20	-
Numboxes		2	2	1.5

Numboxes. A numbox displays a digit on its face and is contained inside a numbox zone (Figure 4). Users can shift their decimal place by grabbing and manipulating the numbox, relocating it relative to the others. This action alters the overall represented number. For instance, when a numbox is moved to a numbox zone positioned to the right of its original placement, all intervening numboxes shift left. Conversely, when a numbox moves left of its original position, intervening numboxes shift right. To illustrate, consider the example of starting with the number 1.2345. By grabbing the numbox containing the decimal point and placing it where the numbox containing the digit 4 is, the overall number changes to 1234.5. This single action effectively alters the order of magnitude of the number by 3. The numbox mechanics also allow users to delete a specific digit by grabbing its numbox and releasing it beyond a small threshold from its original position, as long as it does not overlap with another numbox zone. This action effectively "deletes" the numbox. For example, removing the digit 3 from 1.2345 using this method results in 1.245.

Combination Dial Mechanics. To replicate the mechanics of real-world combination dials, two arrow buttons were integrated into each numbox, positioned above and below the numeric display to function as increment and decrement controls. This design allows users to adjust the value represented in a specific numbox by interacting with the corresponding buttons. (Figure 4).

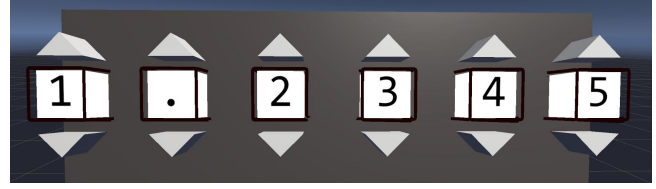


Figure 4: Number display showcasing the value 1.2345 with numboxes (white cubes with black edges) accompanied by decrement (arrow underneath a numbox) and increment (arrow above a numbox) buttons.

Multi-Column Layout. Drawing inspiration from vintage mechanical calculators, we attach a column of 11 buttons to each decimal place, encompassing digits from 9 at the top to 0, along with a decimal point button at the bottom (Fig. 1). By providing the redundancy of a column for each decimal place, we expect users to effortlessly make precise and quick editions of specific digits inside a number.

Delete Button. Additionally, all numpads have a delete button that offer the basic function to delete the rightmost digit, except for the Conventional where due to its explicit cursor, the delete button deletes the digit to the left of the cursor.

Decimal Point. We only permit one decimal point to be present at any given time, as multiple decimal points would render the number invalid. Consequently, if a user wishes to insert a decimal point and one already exists within the number, the existing decimal point must first be deleted before the new one can be inserted.

3.2 Conventional

Our study will use the Conventional numpad as a baseline. We considered a numpad with digits running from 0 to 9, a decimal point, and a delete key. For the layout, we chose the numeric keypad configuration featuring the 7-8-9 buttons two rows above the 1-2-3 buttons, which can be found in many calculators and cash registers. Furthermore, this numpad utilizes a touchscreen for displaying numbers with an explicit cursor indicating where the editing is being made, which can be placed by touching the screen. For instance, this functionality can be found in calculator apps available for smartphones and tablets.

3.3 Adjustable Dials - AD

The AD numpad widget preserves the traditional button arrangement of a conventional numpad but differentiates itself by substituting the touch screen with six interactive numboxes and corresponding increment and decrement buttons. Users can input numbers either by repeatedly using the increment and decrement buttons for each numbox until the desired value is displayed, or by selecting a specific numbox, and directly entering the desired number via the number buttons. Users can even perform edits by sliding numboxes (number or decimal point) to other places on the display or by detaching the numbox off the display. Alternatively, users may combine these redundant input methods to achieve the desired outcome.

3.4 Comptometer

This numpad is characterized by the redundant multi-column button layout, each column is accompanied by a corresponding display box above it. However, it is essential to clarify that these boxes serve solely to display digits and to select the desired character (number or decimal point), while lacking the additional interactable features that a numbox has.

3.5 Adjustable Dials & Comptometer - AD&C

The AD&C numpad widget, is the result of incorporating numboxes, combination dial mechanics and multi-column button layout, notably, the junction of these features makes it so there is no need for explicit numbox selection to input a number; instead, the selection is implicit based on the column used.

4 Study Design

4.1 Variables and Hypotheses

The main goal of this study is to assess whether VR numpad designs with redundant input can improve user experience and efficiency in number input and editing tasks, compared to a conventional numpad. To this end, we adopted a within-subject evaluation method focusing on two independent variables, namely, (i) the *Numpad Widget* consisting of four numpad widgets designated as “Conventional”, “AD”, “Comptometer”, and “AD&C” (Figure 1); and the (ii) *Starter and Target Number Difference* where we considered four distinct Damerau-Levenshtein distances (DLD), each with three examples to avoid participant mental fatigue that could skew the results. We opted to use the DLD as the metric to quantify the similarity between two different numbers as defined by Damerau [13] and Levenshtein [36]. Table 2 lists the pairs of starter numbers (displayed in the numpads) and the target numbers (which the participants were asked to change the starter numbers to) as well as their corresponding DLD distances.

Table 2: Pairs of starter and target numbers with their corresponding Damerau-Levenshtein distances (DLD) assigned to participants per numpad.

Starter number	Target number	DLD
0.0000	0.0001	1
9078.1	907.81	
3.1415	3.1495	
20.400	27.301	3
3783.9	3.8739	
0.8642	60.843	
51.209	21.357	4
129.73	952.83	
66.250	6015.0	
3.3512	111.99	6
123.45	54.123	
75.411	3040.6	

We measured the impact of the independent variables on several dependent variables. In terms of objective measures, we assessed the task completion time required to perform the editions to change

the starter number into the target number (each with different DLD values). Additionally, we considered subjective measures such as user’s perceived system usability, user preference, user satisfaction and perceived cognitive workload. Given these variables, we posited the following hypotheses:

H1. The Conventional numpad will have the longest average task completion time for all DLD tasks.

H2. The Conventional numpad will have the lowest usability, user preference and highest mental workload of all numpad widgets.

H3. AD will decrease the tasks completion time for small edition tasks (DLD1 and DLD3).

H4. The AD&C numpad will have the highest usability, user preference and lowest mental workload of all numpad widgets.

H5. AD&C and Comptometer will have better average task completion time values for all DLD tasks when compared to the Conventional numpad.

4.2 Participants

A total of 30 volunteer participants were involved in the user study, with an average age of 25.8 ± 6.1 years old, comprising 17 males and 13 females. In terms of educational background, 15 participants held Master’s degrees, 9 had Bachelor’s degrees, 4 had completed high school, and 2 had earned PhDs. Regarding VR experience, 13 participants had never used VR, 5 had used it once, 9 used it rarely, 2 used it monthly, and 1 used it daily. Among those who had used VR, the majority utilized the Oculus Quest/Quest 2. Only 5 participants had experience editing/inputting whole numbers in VR, and none had ever performed the task of editing/inputting decimal numbers in VR.

4.3 Apparatus

Participants conducted the experiment in a standing position, utilizing the Meta Oculus Quest 1 head-mounted display with a resolution of 1440×1600 per eye and a refresh rate of 72 Hz, along with Oculus Touch controllers for interacting with the numpads. A laptop computer running Windows 11 Home, equipped with an Intel Core i7-1165G7 processor clocked at 2.80 GHz, an integrated GPU Intel(R) Iris(R) Xe Graphics and 8GB of RAM, was utilized to showcase the tutorial video before the experiment, to record participants sessions and to fill-in post-experiment questionnaires.

The numpad widgets were developed in Unity 2021.3.22f1, utilizing TextMeshPro, Oculus XR Plugin, and XR Plugin Management packages. We used the “VR Button” [53] asset for the buttons and Blender 3.4.0 software to model the digits on the buttons, while interaction functionalities were implemented using the Oculus SDK and C# within Visual Studio Code 1.86.0.

4.4 Tasks

Video Tutorial. We created a short video tutorial covering the functionality of the four numpad widgets, providing participants with an overview of the numpads they will use and instructions on the operations each numpad supports, as well as guidance on how to execute these operations.

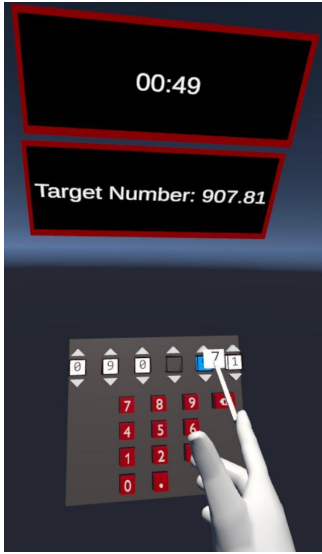


Figure 5: Example of the main task environment where a user is grabbing a numbox while interacting with the AD numpad.

Habituation Task. During the habituation task, participants were positioned in the center of a circle containing the four numpad widgets, each initially displaying the number “1.2345” and featuring different example target numbers above them. Their objective is to input these target numbers on each numpad while minimizing the number of edits/inputs and familiarizing themselves with the numpads’ features. Participants had the freedom to interact with the numpads in any order and for as long as they wish until they feel comfortable. They were also encouraged to ask questions that aroused during the task duration. Whenever participants were satisfied, they could press a ‘start button’ to begin the main task.

Main Task. During the main task, for each numpad, participants were tasked with editing or inputting the twelve starter numbers to match the corresponding target numbers presented in Table 2. Thus, throughout the entire session, each participant performed 48 (3 numbers per DLD x 4 DLD x 4 numpads) number editing/inputting tasks. Upon accessing the task interface, users encounter a numpad widget accompanied by two panels positioned above it (Fig. 5). The first panel displays the target number, while the second panel shows the elapsed time since the beginning of the current task. When the currently input number matches the target number, the timer in the time panel halts, displaying the completion time of the task. Simultaneously, the target panel indicates task completion with a message and initiates a countdown to the next task. Upon completion of the countdown, a new target number is presented alongside a refreshed numpad widget to the participant, resetting the time displayed on the time panel. The sequence of tasks was ordered via Latin squares’ permutations taking into account the numpad widget type and DLD level, where for each level the sequence of starter-target number pairs follows the order outlined in Table 2.

4.5 Procedure

The user study commences with participants completing a consent and demographic questionnaire. Following this, they receive a concise explanation regarding the study’s purpose and the tasks they will undertake. Subsequently, participants watch a video tutorial outlining the procedures. They are then instructed to wear the Oculus Quest and engage in the habituation task until they feel sufficiently acquainted with the numpads (no longer than 3 minutes). Once ready, participants are instructed about the primary objective of the main task: to edit or input the target numbers as swiftly as possible. The main task then commences. Throughout the execution of the main task, users did not receive any explicit instructions to prevent potential influence on their results. All tasks in VR were conducted with participants standing, with no movement required other than using their head, arms, and hands.

Upon completing the main task, participants were asked to fill out three questionnaires: System Usability Scale (SUS)[7], raw NASA Task Load Index (NASA-TLX)[24], and a user satisfaction survey featuring six Likert-scale questions (ranging from 1, “Strongly Disagree,” to 5, “Strongly Agree”) for each numpad widget. Additionally, a user preference survey with eight questions was administered to determine the preferred numpad for each question. Finally, semi-structured interviews were conducted to gather feedback on participants’ interactions with the numpad widgets.

5 Results

5.1 Performance

A two-way repeated measures ANOVA was conducted to determine the effect of different virtual numpad interfaces on the time taken to edit pairs of decimal numbers with different DLDs. Analysis of the studentized residuals showed that there were three outliers, which had a studentized residual value of 4.77, 3.59, and 3.22. Time taken to perform the editing tasks was normally distributed except for the tasks DLD1 and DLD6 of Conventional ($p = .009$; $p = .028$), DLD1 and DLD3 of AD ($p = .001$), DLD1 of Comptometer ($p = .004$), and DLD3 of AD&C ($p = .032$), as assessed by Shapiro-Wilk’s test of normality on the studentized residuals ($p > .05$). Mauchly’s test of sphericity indicated that the assumption of sphericity had been violated for the two-way interaction, $\chi^2 = 63.205$, $p = .033$. Therefore, a Greenhouse-Geisser correction was applied ($\epsilon = .673$). There was a statistically significant interaction between numpad widget used to perform the tasks and DLD level of said tasks on mean time taken to perform the tasks, $F(6.059, 175.715) = 28.349$, $p < .001$. Therefore, simple main effects were examined, and Bonferroni-corrected pairwise t-tests were subsequently conducted to precisely identify the specific condition differences driving the interaction effect.

DLD1. According to Mauchly’s test of sphericity, the assumption of sphericity was largely met ($W = 0.874$, $\chi^2(5) = 7.245$, $p = 0.032$). As a result, the Greenhouse-Geisser correction was not strictly necessary, but the epsilon value was $\epsilon = 0.874$. The analysis revealed a statistically significant interaction between the numpad widget and the performance of DLD1 tasks, $F(3, 87) = 4.886$, $p = 0.003$, $\eta_p^2 = 0.102$, which indicates a medium effect of the different conditions on DLD1 values.

Table 3: Pairwise comparison of the mean difference in task completion for each numpad by DLD level. Mean time in seconds. Rows highlighted in bold indicate statistically significant differences in the pairwise comparison. Negative mean values refer to less time taken between the corresponding numpads pairwise comparison.

DLD	Pairwise comparison	Δ Mean time [s]	p-value
1	AD vs. Conventional	-1.500	.298
	Comptometer vs Conventional	-1.790	.028
	AD&C vs Conventional	-2.160	.002
	Comptometer vs AD	-0.289	1
	Comptometer vs AD&C	0.367	1
	AD&C vs AD	-0.656	1
3	AD vs. Conventional	-2.800	.061
	Comptometer vs Conventional	-2.870	.021
	AD&C vs Conventional	-4.690	<.001
	Comptometer vs AD	-0.070	1
	Comptometer vs AD&C	1.820	.005
	AD&C vs AD	-1.890	.065
4	AD vs. Conventional	1.480	.067
	Comptometer vs Conventional	-3.160	<.001
	AD&C vs Conventional	-1.700	.034
	Comptometer vs AD	-4.630	<.001
	Comptometer vs AD&C	-1.460	.016
	AD&C vs AD	-3.180	<.001
6	AD vs Conventional	8.760	<.001
	Comptometer vs Conventional	0.467	1
	AD&C vs Conventional	1.630	.051
	Comptometer vs AD	-8.29	.075
	Comptometer vs AD&C	-1.170	.005
	AD&C vs AD	-7.120	<.001

DLD3. Mauchly's test of sphericity revealed a violation of the assumption ($W = 0.507$, $\chi^2(5) = 18.835$, $p = 0.002$). As a result, the Greenhouse–Geisser correction was applied ($\epsilon = 0.769$). The analysis indicated a statistically significant interaction between the numpad widget and the execution of DLD3 tasks, $F(2.307, 66.909) = 11.498$, $p < 0.001$, $\eta_p^2 = 0.212$, which indicates a large effect of the different conditions on DLD3 values.

DLD4. For DLD4, Mauchly's test indicated that the assumption of sphericity was satisfied ($W = 0.880$, $\chi^2(5) = 7.245$, $p = 0.203$). The analysis revealed a statistically significant interaction was detected between the numpad widget and the performance of DLD4 tasks, $F(3, 87) = 31.341$, $p < 0.001$, $\eta_p^2 = 0.309$, which indicates a large effect of the different conditions on DLD4 values.

DLD6. Mauchly's test indicated a breach of the sphericity assumption ($W = 0.644$, $\chi^2(5) = 12.199$, $p = 0.032$). Consequently, the Greenhouse–Geisser correction was applied ($\epsilon = 0.786$). The analysis revealed a statistically significant main effect of *Condition* on DLD6 tasks, $F(2.358, 68.387) = 85.429$, $p < 0.001$, $\eta_p^2 = 0.574$, which indicates a very large effect of the different conditions on DLD6 values.

Comparisons. The mean task completion time, dispersion, and skewness per task DLD for the four numpad widgets are illustrated

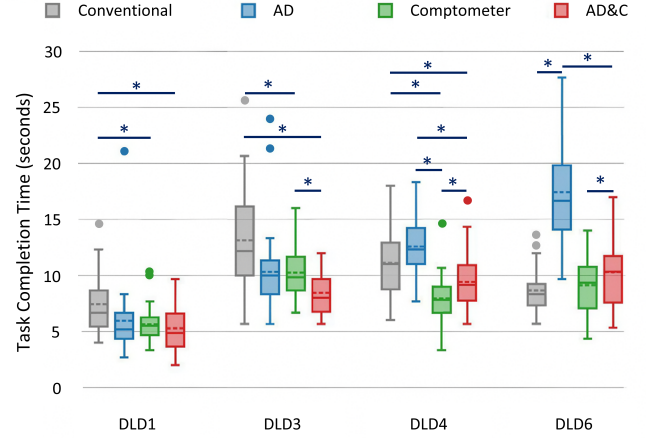


Figure 6: Mean task completion time in seconds per task Damerau-Levenshtein distances (DLD) for the four numpad widgets. DLD quantifies the number of edits required to transform one number into a target number.

in Figure 6. Table 3 shows the pairwise comparison between the four numpad widgets for each DLD task.

5.2 Qualitative Metrics

System Usability Scale. Among the numpad widgets, the Comptometer widget had the highest mean SUS score (82.4 ± 14), closely followed by the AD&C widget (81.3 ± 15.2). The Conventional widget scored 76.9 ± 16.5 , while the AD widget had a significantly lower score (69 ± 24.9). A Friedman test was conducted to determine if there were differences in SUS score evaluations among the four numpads. The test indicated that there were statistically significant differences in SUS scores across the four numpads, $\chi^2(3) = 9.690$, $p = 0.021$. Post hoc pairwise comparisons revealed that SUS scores were significantly different when comparing AD with AD&C ($p = 0.011$) and AD with Comptometer ($p = 0.009$), with statistical significance accepted at $p < 0.0167$.

NASA Task Load Index. The mean and standard deviation of NASA-TLX scores for the Conventional, AD, Comptometer, and AD&C numpads were (28.64 ± 21.63), (28.56 ± 21.46), (25.39 ± 18.08) and (20.97 ± 17.08), respectively. A Friedman test was conducted to determine whether there were significant differences in NASA-TLX evaluations among the four numpads. The results indicated no statistically significant differences, $\chi^2(3) = 6.116$, $p = 0.106$.

User Satisfaction. Results pertaining to user satisfaction questions (Q1 to Q6) are illustrated in percentile form in Figure 7. For Q1, “**This widget is easy to use**” the mean satisfaction scores for the Conventional, AD, Comptometer, and AD&C numpads were 4 ± 1.1 , 3.67 ± 1.1 , 4.43 ± 0.63 , and 4.43 ± 0.68 , respectively. A Friedman test indicated significant differences among the numpads, $\chi^2(3) = 11.356$, $p = 0.01$. Post hoc pairwise comparisons revealed a significant difference between AD and AD&C ($p = 0.009$).

Regarding Q2, “**This widget is fun to use**” mean scores were 2.8 ± 1.27 , 3.93 ± 1.34 , 3.73 ± 1.11 , and 4.63 ± 0.72 for the Conventional,

AD, Comptometer, and AD&C numpads, respectively. The Friedman test showed significant differences, $\chi^2(3) = 34.819, p < 0.001$. Post hoc analysis found significant differences between Conventional and AD ($p < 0.001$), Conventional and AD&C ($p < 0.001$), and Comptometer and AD&C ($p = 0.009$).

Concerning Q3, “I was able to learn how to use this widget easily” mean scores were 4.53 ± 0.68 , 4.07 ± 1.01 , 4.87 ± 0.35 , and 4.53 ± 0.57 for the Conventional, AD, Comptometer, and AD&C numpads, respectively. The Friedman test revealed significant differences, $\chi^2(3) = 18.322, p < 0.001$, with a significant difference between AD and Comptometer ($p = 0.001$).

For Q4, “I was able to quickly input the target number using this widget” mean scores were 3.4 ± 1.3 , 3.67 ± 1.15 , 4.1 ± 0.88 , and 4.43 ± 0.82 for the Conventional, AD, Comptometer, and AD&C numpads, respectively. The Friedman test indicated significant differences, $\chi^2(3) = 12.678, p = 0.005$. Post hoc analysis showed a significant difference between Conventional and AD&C ($p = 0.003$).

In terms of Q5, “I was able to accurately input the target number using this widget” mean scores were 4.23 ± 0.73 , 4.17 ± 1.05 , 4.33 ± 0.66 , and 4.5 ± 0.68 for the Conventional, AD, Comptometer, and AD&C numpads, respectively. The Friedman test indicated no significant differences, $\chi^2(3) = 1.682, p = 0.641$.

As for Q6, “This widget required little physical effort to use” mean scores were 3.6 ± 1.54 , 3.67 ± 1.32 , 3.8 ± 1.19 , and 3.77 ± 1.17 for the Conventional, AD, Comptometer, and AD&C numpads, respectively. The Friedman test indicated no significant differences, $\chi^2(3) = 0.662, p = 0.882$.

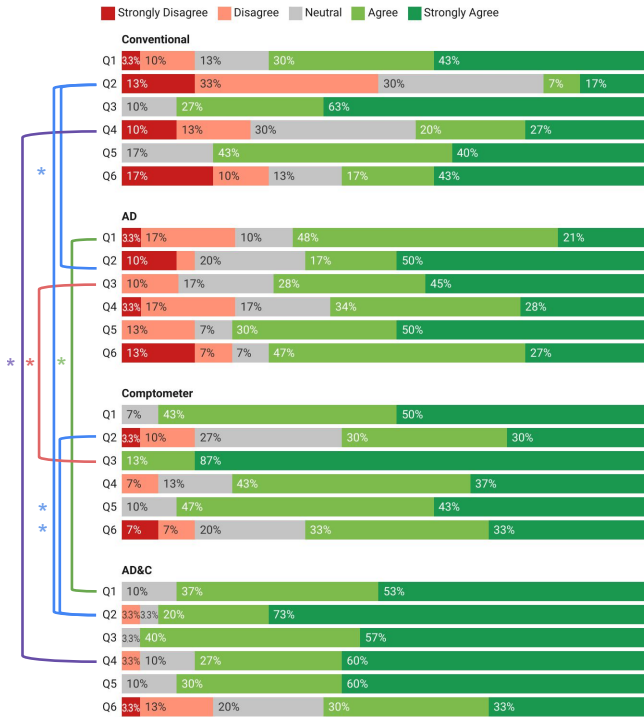


Figure 7: Users satisfaction with each numpad, assessed using Q1–Q6. Vertical line segments topped with an asterisk (*) denote statistically significant difference.

User Preference. A total of eight questions (Q7 to Q14) were posed to evaluate participants’ preferences for the different numpads.

For Q7, “Which widget is easiest to use?”, 50% of participants chose AD&C, followed by 26.6% favoring the Comptometer, 13.3% for the Conventional numpad, and 10% selecting AD.

In response to Q8, “Which widget was more fun to use?”, the AD&C received the highest preference at 53.3%, with 30% choosing AD, 10% selecting the Comptometer, and 6.6% favoring the Conventional numpad.

Regarding Q9, “Which widget is easiest to learn?”, the Conventional numpad was most preferred with 60%, while the Comptometer was chosen by 30%, AD&C by 6.6%, and AD by only 3.3%.

For Q10, “Which widget allowed for quicker input of the target number?”, the AD&C again ranked highest with 60%, followed by both AD and the Comptometer with 16.6% each, and 6.6% for the Conventional numpad.

In response to Q11, “Which widget allowed for more accurate input of the target number?”, 43.3% of participants preferred AD&C, while 30% selected the Conventional numpad, 16.6% chose AD, and 10% opted for the Comptometer.

For Q12, “Which widget required less physical effort to use?”, the Conventional numpad and the Comptometer tied at 30% each, with AD at 20% and AD&C also at 20%.

Regarding Q13, “Which widget was more reliable?”, the Conventional numpad and the Comptometer both received 30%, with AD and AD&C each receiving 20%.

Finally, in response to Q14, “Which widget did you prefer overall?”, the majority of participants (60%) favored AD&C, followed by AD with 16.6%, the Comptometer with 13.3%, and the Conventional numpad at 10%.

6 Discussion

The present study assessed how redundant interfaces can enhance user experience and efficiency when it comes to precise number editing of floating point values in VR environments. The proposed redundant interfaces offer specialized input techniques tailored for precise numerical editing in immersive environments, enabling direct manipulation of decimal places, reducing input errors, and improving efficiency.

Although we focus on a very specific task, it remains highly relevant as Spatial Computing continues to integrate more deeply into everyday productivity and routine activities that require precise floating point number input or editing. For instance, VR/AR/MR floating-point value editing is crucial in applications that demand numeric precision, such as CAD modeling (e.g., dimensioning the height of a window or the radius of a car wheel) [37, 74], money transaction (e.g., entering prices and investment values with high accuracy) [31], mathematical visualization (e.g., entering shape parameters of an implicit surface function) [10], scientific simulations (e.g., to change mass of objects while solving the Equations of Motion) [20, 48], flight simulators (e.g., precise numerical inputs for damage values or altitude settings) [64], smart home or IoT simulators (e.g., adjust parameters like temperature and energy consumption) [30], medical training (e.g., precise dosage of an anesthetic) [39] or rehabilitation to access functional performance (e.g.,

dual-task activities that combine motor and cognitive tasks such as inputting a number) [18].

Unlike whole-number inputs, floating-point values require fine-tuned modifications, making traditional numpad designs less efficient for such tasks. Conventional VR numpads, initially designed for discrete number entry (e.g., PINs or ZIP codes) [44], provide limited interaction techniques: they typically confine users to typing numbers via buttons and, in rare cases, allow only basic edits by positioning a caret/cursor at a specific decimal place. More often, they require users to re-enter numbers from scratch, introducing unnecessary cognitive and motor effort.

By moving beyond WIMP-based paradigms, this work fills a gap in VR interface design, offering specialized input techniques tailored for precise numerical editing in immersive environments. To enhance usability, reduce input errors, and improve efficiency, we incorporate several input methods or shortcuts [73] such as adjustable dials and numboxes. These redundant inputs make numerical entry more natural and adaptable to diverse tasks and user needs. Furthermore, this approach aligns with best practices and principles for VR interaction design, ensuring a more user-friendly and effective experience [28].

Overall, the redesigned numpads that incorporate redundancy were well-received by participants, providing clearer and more efficient input for floating-point values. While no single numpad consistently outperformed across all metrics, the results suggest that numpads with redundant elements offer advantages depending on the specific DLD level.

User feedback also correlated with the measured variables. For instance, based on the semi-structured interviews, most participants viewed the Comptometer, AD&C, and AD widgets as viable alternatives to the conventional method for decimal input. However, participants noted that the choice of widget often depends on the specific task (i.e., DLD level). The Comptometer and AD&C widgets were praised for their multi-column redundant layout, which allowed for more efficient number input without needing to delete or reposition the cursor. Both the AD&C and AD widgets also featured digit movement and buttons for incrementing or decrementing numbers, which participants found useful for small precise adjustments. The AD&C widget stood out for its high interactivity and extensive toolset, while the AD was valued for its compact design, making it efficient for editing numbers in VR.

We attribute the improved numerical editing efficiency of our proposed widgets primarily to the ability for users to directly modify any decimal place of a number. This is accomplished via redundancy input: either by interacting with increment/decrement buttons (in the AD), selecting a specific column (as in the Comptometer) or using both methods simultaneously (in the AD&C model). In contrast, conventional numpads require users to position a caret/cursor to specify which decimal place they wish to edit, introducing an extra step and reducing overall efficiency.

In general, user performance when inputting or editing floating-point numbers on numpads is influenced by both the features of the numpad and the complexity of the tasks. For simpler tasks (DLD1 to DLD3), all redesigned numpads outperformed the conventional numpad. This suggests that features like numboxes, dials, and multi-column layouts help reduce input errors by offering more flexible interaction methods. One reason for this improvement is

that the conventional numpad, which relies on button presses and explicit cursor placement, often requires additional corrections. However, for more demanding tasks (DLD4 to DLD6), the conventional numpad proved to be the most effective, suggesting that in more complex tasks, direct number entry is more efficient than managing multiple interactive elements. These results may be linked to users' lack of familiarity with the redundant interaction techniques, which could hinder performance in tasks with larger DLD values.

Concerning **H1**, this hypothesis was validated as the numpads incorporating numboxes, specifically the AD&C and AD, demonstrated significantly lower mean task completion times compared to the Conventional numpad for DLD1 and DLD3 tasks. This finding suggests that numboxes facilitate more efficient execution of small editing tasks.

Our prediction for **H2** was not supported, as the AD widget had poorer usability scores and similar mental workload results compared to the other numpads. However, the Conventional was the least preferred numpad overall.

Regarding **H3**, we accepted this hypothesis for DLD1 and DLD3 tasks. However, for DLD4 and DLD6 tasks, the AD exhibited the highest mean task completion time. This increased time can be attributed to the high number of selections required to interact with the widget. Specifically, users needed to select the active numbox before they could enter a number, making the interaction more time-consuming as the DLD levels increased.

H4 was partially confirmed since the AD&C numpad had the lowest mental workload and was significantly preferred. Although the Comptometer numpad had a slightly better usability score than the AD&C, its score was still higher than the Conventional numpad.

Similarly, **H5** was supported for DLD1, DLD3, and DLD4 tasks. At the DLD6 level, the performance of the Conventional numpad equaled that of the Comptometer numpad and, although not statistically significant, showed a considerable advantage over the AD&C widget. Contrary to our expectations, the efficiency of the Conventional numpad increased with higher DLD levels. This unexpected result was due to users in DLD1 and DLD3 tasks struggling to place the cursor accurately on numbers that required editing, owing to the proximity of numbers on the touchscreen. As the DLD levels increased, the higher number of required edits made it more efficient for users to delete everything and start inputting from scratch.

The proposed numpad designs take advantage of VR's spatial freedom, utilizing the widely available virtual "real estate" to incorporate interface redundancy. While such an approach would be restrictive in physical environments, VR allows for larger numpads that trade extra space for improved input clarity. Our findings can guide future VR interface designs for symbolic data input [28]. In particular, our findings suggest a *design guideline* for selecting numpad layouts related to DLD complexity: for tasks requiring multiple inputs or frequent edits, the conventional numpad proves to be the most efficient; conversely, for tasks involving fewer inputs or minimal edits, the Comptometer numpad offers the best efficiency. Our findings also point to a *design guideline* related to spatial constraints within the virtual environment: since the Comptometer and AD&C numpads occupy significantly more space than the Conventional and AD numpads, developers can opt for more compact designs in scenarios where available space is limited or cluttered

with other virtual elements. In such cases, the AD numpad offers a strong compromise, outperforming the Conventional numpad for tasks with frequent edits while remaining more engaging to use and more space-efficient than VR numpads with larger layouts such as the Comptometer and AD&C.

7 Limitations & Future Work

Despite the positive results, we acknowledge several limitations that should be addressed in future studies on symbolic data input, particularly, in the design of VR/AR/MR numpads.

First, we did not collect sufficiently detailed data to stratify performance by VR experience level, and thus our findings mainly reflect novice VR users. Future work should include expert populations (e.g., CAD or 3D modeling professionals) to determine applicability in real-world workflows.

Second, our results stem from a relatively low spatial complexity environment—one with minimal virtual element density, low intricacy, and no significant occlusions. Although larger panels enhance clarity for precise number input, they cause occlusion of other virtual content and require the use of larger muscle groups (shoulder, elbow, and wrist), potentially leading to gorilla arm effects (i.e., upper limb fatigue) and reduced user comfort during extended use [28]. To mitigate these issues, careful consideration must be given to the size and placement of numpads in relation to surrounding elements, and whether the numpad will temporarily occlude other content. A possible solution is dynamic resizing—where the interface enlarges only upon selection and reverts once the input task is completed. Alternatively, the numpad could temporarily shift to a side position during editing, ensuring minimal disruption to the overall scene while maintaining an efficient workflow. These approaches ensure that the scene is only temporarily affected, and since the user is focused on editing the number, the rest of the scene becomes less important during this process.

Regarding scalability, this study focused on six-digit floating-point values, while the majority of XR applications typically require the entry of whole-numbers, such as PINs, credit card numbers, ZIP codes or birth dates. Yet, switching floating-point values to whole-numbers is straight-forward: the decimal point just needs to be replaced by an extra digit. However, adding more decimal places can pose a challenge, as the Comptometer and AD&C designs scale poorly for larger numbers, hence, limiting the proposed numpads' ability to handle higher digit counts. We consider this a less critical limitation since most VR applications do not require number precision beyond six or seven digits, even in 3D content creation or measurement apps numbers rarely go beyond this range. Other limitations our study consist of narrowing the tasks to number edition. Future numpad design iterations could explore the input of numbers from scratch or even having the user perform a calculation (e.g., addition, multiplication), though this would require a much larger sample size and significantly increase study complexity.

A further limitation is that our evaluation relied on task completion time and DLD as primary metrics, but additional objective measures such as input accuracy (e.g., number of wrong entries, misselections, undo operations) and physical effort (e.g., hand travel distance) would provide deeper insights into user behavior and enable time–accuracy trade-off analyses. Our study did not capture

fine-grained logs of feature-level usage, thus limiting our ability to determine which design elements contributed most to performance and preference (e.g., which redundant elements of the AD&C were used most).

We also acknowledge that our study focused exclusively on controller-based raycasting, one of the most widely adopted and reliable interaction methods in current VR systems. Such allowed us to systematically compare redundant numpad designs without the variability introduced by multimodal input alternatives. Hence, our results should be interpreted as advocating redundancy-based numpads not as replacements but as complementary techniques that may integrate well into multimodal workflows (e.g., passthrough physical keyboards, speech recognition, gaze-based input, direct hand-tracking keyboards). For example, coarse input may be more efficiently handled by speech or gesture, while redundant numpads provide clear benefits when floating-point precision, frequent corrections, or direct editing are required. A promising direction for future research is to explore hybrid workflows that dynamically combine input modalities, allowing users to fluidly switch between input channels depending on task demands.

Finally, our study focused on floating-point number input and editing but omitted broader workflows typical of real applications, which involve locating input fields, alternating between number editing and other tasks (e.g., 3D modeling), and managing cognitive context switches. As a result, the real-world benefits of redundant numpads remain to be addressed in more complex and ecologically valid scenarios, where task switching may introduce additional challenges.

8 Conclusions

In this study, we conducted a user study to compare the effect of redundant interfaces on numpads to input floating point values inside a VR environment. Our key finding is that redundancy input assists the user to perform better when compared to the conventional VR numpad, which remains the standard despite its limitations. In particular, our results indicate that a multi-column layout improves time completion task for small number edits more than the numbox feature alone, yet, the combination of both these interaction elements significantly enhances user interaction. Moreover, results from the comparative study suggest a guideline for selecting numpad layouts based on DLD complexity. For tasks requiring multiple inputs or frequent edits, the conventional numpad remains the most efficient. Whereas, for tasks involving fewer inputs or minimal edits, the Comptometer numpad offers the best efficiency.

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Figure Credits

Figure 3 shows images of a Comptometer [50] and a combination lock [65], both of which belong to the public domain.

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